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PALEOMAGNETIC RESULTS FROM MALLORCA (BALEARIC ISLANDS, SPAIN)

R. Freeman,[1] F. Sàbat,[2] W. Lowrie,[1] and J.-M. Fontboté[2]

Abstract. We studied the paleomagnetism of Jurassic limestones on the island of Mallorca (Balearic Archipelago, western Mediterranean) to determine if two thrust sheets exhibiting different structural patterns were rotated relative to one another. After correction for tilt the circle of confidence for the mean paleomagnetic direction from sites in areas associated with frontal ramps overlaps with the confidence circle of the mean for those sites associated with oblique-lateral ramps. This indicates that there has been no differential rotation in spite of the fact that the fold axes in these areas trend almost perpendicular to one another. Although the age of magnetization is prefolding (i.e., before Oligo-Miocene), it is not certain that the paleomagnetic directions reflect Jurassic magnetization since they may have been acquired during a period of emersion in the Upper Cretaceous or Paleocene. The resulting mean direction for nine sites from both thrust sheets (declination = 41°, inclination = 48°, $\alpha 95$ = 7°) implies significant clockwise rotation of the whole island, or at least of the Mesozoic cover, since acquisition of magnetization. If part of this rotation occurred, as we believe, in the Neogene, then structural data from pre-Miocene rocks must be corrected before being used for regional tectonic interpretation. The mechanism to produce 40° of clockwise rotation since the Oligo-Miocene could be combined regional extension, Alpine compression, and block rotation.

[1]Institut für Geophysik, ETH - Hönggerberg, Zürich, Switzerland.

[2]Departament de Geologia Dinamica, Geofisica i Paleontologia, Facultat de Geologia, Zona Universitaria de Pedralbes, Barcelona, Spain.

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THE STRUCTURAL PROBLEM

Paleomagnetic declinations are sensitive to rotations about a vertical axis and can therefore help quantify differential rotations and constrain kinematic models. They can aid in understanding the kinematics of both thrust- and wrench-related tectonics. With this goal in mind we applied the paleomagnetic method to a structural problem on the island of Mallorca.

The largest of the Balearic Islands (Figure 1), Mallorca is bounded by two imbricate piles of thrust sheets, the Serra de Tramuntana along the NW coast, and the Serres de Llevant along the SE coast. Although the strikes of most thrusts and fold axes are oriented NE-SW, an important region of the Serres de Llevant exhibits thrust strikes and fold axes oriented NW-SE, that is, nearly perpendicular to those of the Serra de Tramuntana (Figure 2). The aim of the present study is to compare paleomagnetic directions from two thrust sheets which show very different structural patterns to help determine how these structures acquired their contrasting orientations.

Serra de Tramuntana

Earlier work by Fallot [1922] distinguished three main tectonic units in the Serra de Tramuntana which he designated Series I, II, and III (Figure 2). More recent work [Del Olmo and Alvaro, 1984] further subdivided these into smaller units. All these units are allochthonous thrust sheets. The lowermost unit (Series I of Fallot [1922]) were, until recently, considered autochthonous. Several meter scale tectonically intercalated lenses (horses) of Paleozoic rocks have been found within the Lower Triassic red beds at the bottom of Series I [Ramos and

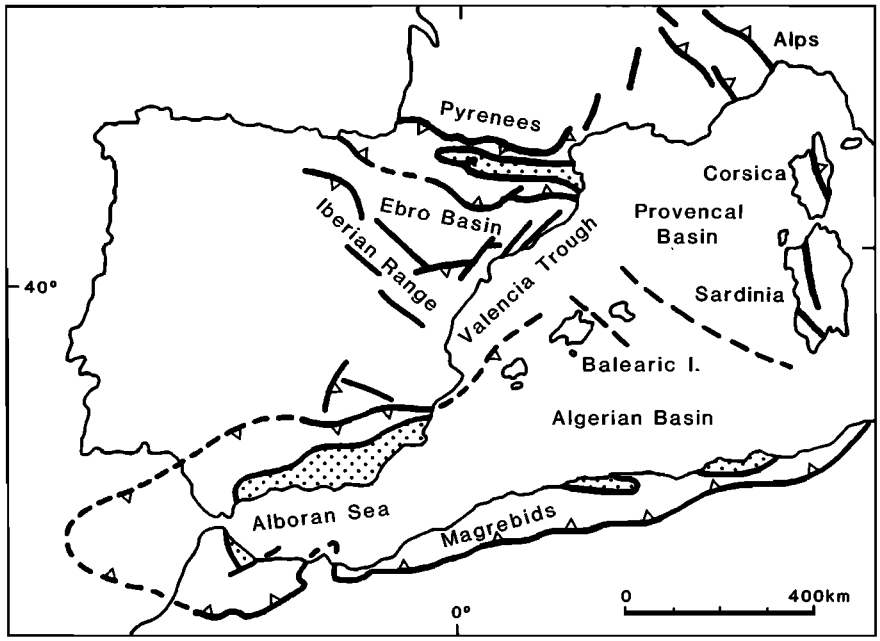


Fig. 1. Location of the Balearic Islands within the western Mediterranean. Mallorca is the largest island of this archipelago. Dotted areas correspond to the Internal Betics, Magrebids, and to the Pyrenean Axial Zone. Dashed lines refer to inferred faults.

Rodriguez-Perea, 1985]. This strongly suggests that Series I is also allochthonous. Further evidence for allochthony is provided by a borehole which cuts 58 m of Middle Triassic (N. Soler, personnal communication, 1986) red clay, gypsum, and thin-bedded dark dolomites located beneath Lower Triassic red beds of Series I [Baron et al., 1985].

In the Serra de Tramuntana thrusts merge and form an imbricated thrust system. Thrusts strike NE-SW and dip SE or are subhorizontal (Figure 2) with transport direction from SE to NW [Fallot, 1922; Alvaro and Del Olmo, 1984]. Several folds can be observed trending NE-SW, plunging subhorizontal, and facing NW. Exposed windows and klippes show a minimum displacement of 7-8 km for several thrusts. Minimum shortening deduced from balanced cross sections is 24 km (55%) [Alvaro, 1987]. The age of major thrusting and associated folding in the Serra de Tramuntana is Langhian (middle Miocene). Some evidence for Oligocene folding exists [Fallot, 1922; Pomar et al., 1983; Alvaro and Del Olmo, 1984], but the significance has not been evaluated.

Serres de Llevant

Darder [1925] described five thrust sheets in the Serres de Llevant. Many thrusts are horizontal; strikes of those with any significant dip do not show consistent orientation. Thrust traces do not exhibit the subparallel pattern which is evident in the the Serra de Tramuntana (Figure 2). Recent structural mapping [Sàbat, 1986] describes thrust sheets which are imbricated with structurally higher sheets found both to the SE and to the SW, the uppermost at the SW edge. This geometry is produced by several thrust sheets which replace each other to the SW, wedging out at lateral ramps. Transport direction is from SE to NW, as deduced from the pattern of younging directions at ramps and the strike of tear faults. Axes of thrust-related folds are subhorizontal, most of them trend either NE-SW, parallel to the frontal ramps, or NW-SE, nearly parallel to the lateral ramps. In general, NE-SW folds face NW; most of the NW-SE folds face NE. Deduced shortening in a SE-NW section across the northern part of the Serres de Llevant is 24 km (55%). However, relative translation between the lowermost and uppermost exposed thrust sheets, calculated by three-dimensional restoration using branch lines, is about 80 km, due to large movement along several oblique-lateral ramps. This movement, and not the NE-SW regional shortening, is responsible for the described SW thrust sheet imbrication. Some thrusts are dated as Oligocene, others as lower and middle Miocene. Thrusting finished by Serravalian (middle Miocene) times [Sàbat et al., 1988].

NW-SE folds in the Serres de Llevant are parallel to and genetically associated with oblique-lateral ramps. Fold axes form a small angle with the thrust transport direction. No major oblique-lateral ramps have been observed in Serra de Tramuntana where NE-SW fold axes are perpendicular to thrust transport direction.

PALEOMAGNETIC RESULTS

A thrust sheet in the Serra de Tramuntana exhibits folds perpendicular to transport direction whereas a thrust sheet of the

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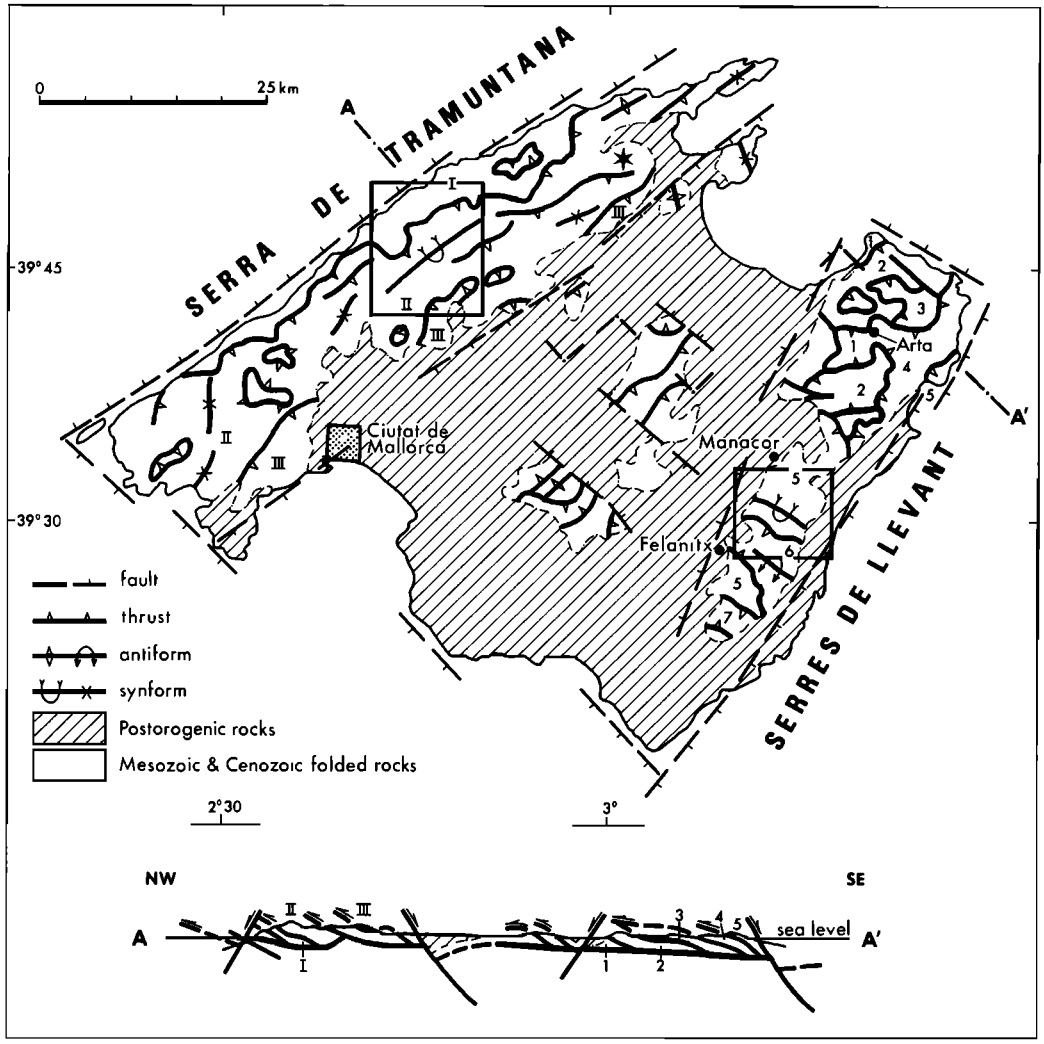


Fig. 2. Structural sketch of Mallorca and general geologic cross section (A-A'). The main geological feature is thrust sheet imbrication affected by late extensional and wrench faults. Outlined squares correspond to areas represented in Figures 3 and 4. I, II, and III in Serra de Tramuntana refer to thrust sheets defined by Fallot [1922], numbers in Serres de Llevant refer to those defined by Sàbat [1986]. Star in NE part of the Serra de Tramuntana corresponds to site MJ06.

Serres de Llevant contains structures subparallel to transport direction (folds and oblique-lateral ramps). We have compared paleomagnetic directions in these two thrust sheets to determine possible rotations related to oblique-lateral structures.

Sampling

Fine-grained pelagic limestones, although not unproblematic, frequently carry stable paleomagnetic directions. On Mallorca, facies with this lithology are represented by Middle and Upper Jurassic pelagic limestones. Most of these indicate little or no dolomitization. Metamorphic effects in the Jurassic limestones are practically nonexistent; ammonites and radiolarians are deformed only in very rare cases and then only minimally. Except where associated with thrust surfaces and in the cores of some folds, no tectonically induced stylolites and no cleavage development appear.

In the Serra de Tramuntana we sampled locations in the tectonic units of Series II of Fallot [1922] because the outcrop situation was most favorable, and the structural relations with NE-SW trending folds were more clearly displayed than the other series. These NE-SW trending folds are almost cylindrical and are perpendicular to thrust transport direction. We sampled six sites, five of which are located close to the core of a NNW facing complex syncline more than 15 km long (see map of locations and cross sections in Figure 3 and descriptions in

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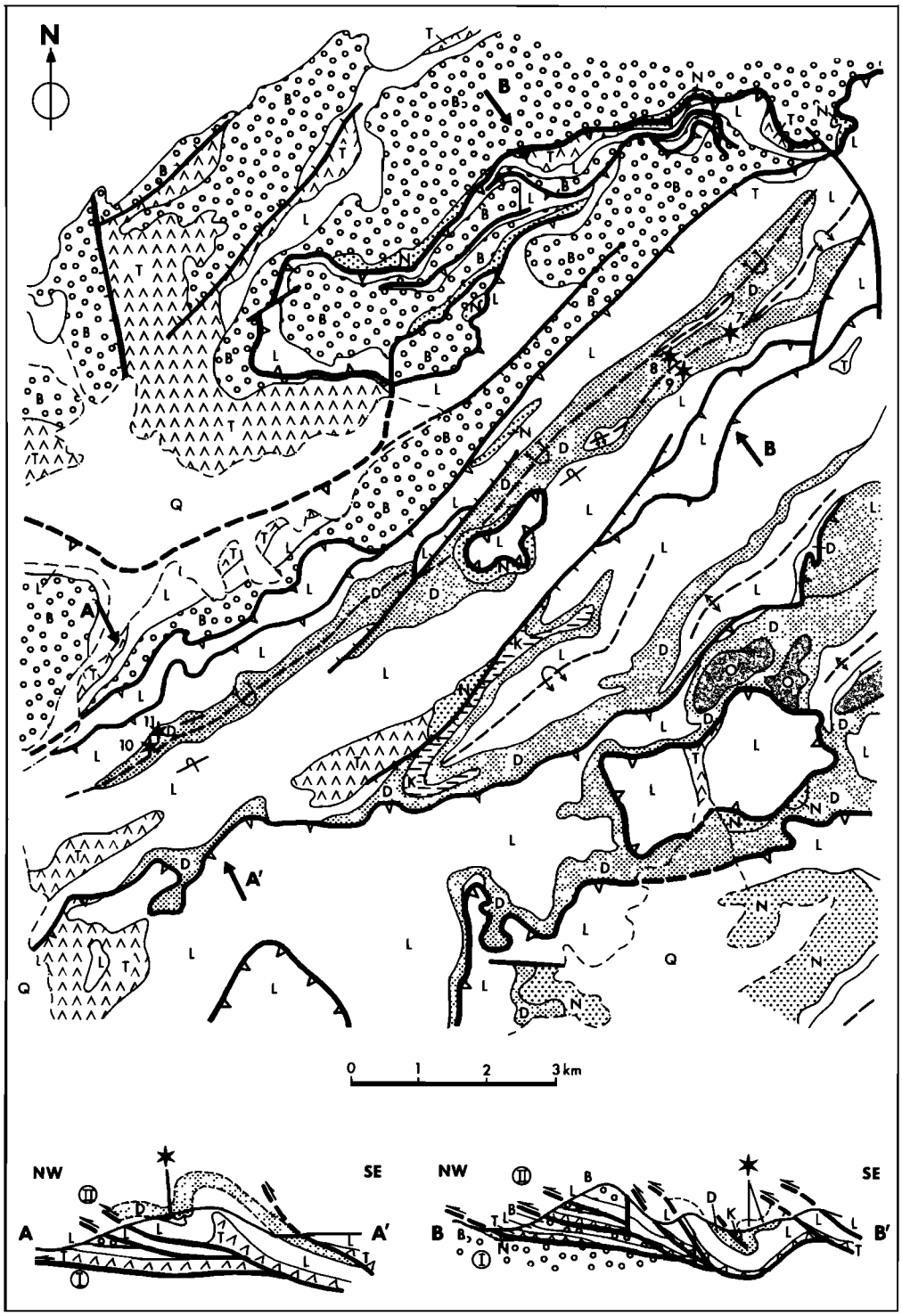


Fig. 3. Paleomagnetic site locations in the Serra de Tramuntana. Geological map of the part outlined in Figure 2 and two partial cross-sections (A-A' and B-B'). Map is simplified from Instituto Geologico y Minero de España (unpublished data, 1986). T, Triassic; L, lower Jurassic; D, middle and upper Jurassic; K, Cretaceous; E, Eocene; O, Oligocene; N, Lower Miocene; Q, Quaternary; B, Breccias (age not accurately known). Paleomagnetic sites are located in the core of a syncline 14 km long and trending NE-SW. Sampled sites are marked by stars (7 is MJ07; 8 is MJ08, 9 is MJ09, 10 is MJ10, 11 is MJ11). In the cross sections, I and II refer to thrust sheets defined by Fallot [1922].

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Table 1). The last site is located farther east of the syncline termination but lies in a structurally equivalent position.

The best outcrops of oblique-lateral structures were found in the southern part of the Serres de Llevant, where we delineated a smaller area south of Manacor (for location see Figure 4). In this area the shortening perpendicular to transport direction is more than other areas and offered the maximum chance of obtaining kinematic information. We sampled seven sites in the normal limb of a recumbent syncline facing to the NE and 6 km long, as this provided the optimum outcrops in the most continuous structure. All sites were chosen as far as possible from local structures to avoid secondary distortions. At each site a minimum of twelve 2.54-cm diameter cores were drilled, and these were later cut into cylindrical samples 2.25 cm in length.

Identification of ferromagnetic mineralogy by isothermal remanent magnetization (IRM) experiments

Acquisition of IRM disclosed that low-, midrange-, and high-coercivity phases were present in significant amounts in most lithologies (Figure 5). In order to identify the ferromagnetic mineral phases, we applied a method due to Lowrie and Heller [1982] which was modified according to a suggestion by F. Heller (personnal communication, 1984). Figure 5 illustrates some features of this experiment which entails three main steps: (1) the "normal" stepwise acquisition of an IRM from 0.03T to 1.0 T with the DC field applied along the cylindrical axis of the sample, (2) application of a single step of 0.1 T in the radial direction (i.e., perpendicular to the 1.0 T component), and (3) stepwise thermal demagnetization to 675°C.

A very rough, semiquantitative estimate of the different contributions making up the IRM can then be calculated according to the following assumptions: (1) the magnetization drop after heating to 150 °C is due only to goethite; (2) the magnetization acquired during the initial IRM acquisition from the natural remanent magnetization (NRM) state to 0.1 T is due to low-coercivity phases such as titanomagnetite or maghemite (as might be expected, this low-coercivity acquisition is in almost all cases nearly equal to the magnetization lost from the axial 1.0 T nearly-saturated IRM when the radial 0.1 T field is applied); (3) the 1.0 T "saturation" magnetization minus the goethite and "low-coercivity" magnetization is due to coarse-grained (specular) and fine-grained (microcrystalline) hematite; and (4) most of the the magnetization left after thermal cleaning to 575°C (Curie temperature of Ti-free magnetite) is due to microcrystalline and specular hematite with high unblocking temperatures.

In the above simple scheme any contribution of highly elongated single-domain magnetite (which could contribute to the saturation magnetization from 0.1 T to 0.3 T) is considered negligible. There was no consistent drop in magnetization in the range 300°C to 400°C (T_n (pyrrhotite) $\leq$ 350°C) nor was there directional changes during NRM demagnetization between 300°C and 400°C. These observations discount any influence of pyhrrotite or maghemite phases to the NRM. In addition, contributions from microcrystalline hematite to the magnetization below 575°C are also assumed to be minor. However, since the unblocking temperatures of microcrystalline hematite are grain size dependent, a small amount of the magnetization lost in the range 150°C to 575°C may be due to not only titanomagnetite, but also fine-grained single-domain microcrystalline hematite.

Referring to Figure 5, it can be seen that low-coercivity phases (low field "inclinations") dominate in sample MJ01.08B below 500°C and high-coercivity hematite phases (high field "inclinations") above 500°C. No evidence for low-coercivity phases can be seen in the acquisition curve for sample MJ14.07C, whereas there is clear evidence for demagnetization of low-coercivity phases between 100°C and 500°C. This is due to the hematite component which overwhelms the acquisition curves and masks the low-coercivity IRM below 0.15 T. Thus magnetization loss isolated between 100°C and 500 °C may reflect the low-coercivity phases even in a sample that from IRM curves appears to be dominated by hematite.

Figure 6 shows the analysis of the two-component IRM experiments applied to all sites of Jurassic Mallorca limestones from which we list paleomagnetic results. This method offers a quick way in which to distinguish lithologies on the basis of their IRM-ferromagnetic mineralogies. Goethite, "magnetite" (which includes all low-coercivity phases), and hematite are present in all lithologies in varying amounts. Low-coercivity "magnetite" phases carry most of the magnetization in all sites except for MJ03, MJ08, and MJ09. Specular hematite is a major contributor in the latter two sites which made up the lithologies which offered the only fold test we found. The paleomagnetic results from these two sites will be discussed in more detail below and were not used in the final statistics.

Thermal demagnetization of the NRM

Because of the presence of high-coercivity phases (goethite and hematite), thermal demagnetization was chosen for routine processing. The goethite component was destroyed by 150°C and carried a present day field (PDF) direction and, as might be expected from the IRM results (Figure 6a), often accounted for up to 90% of the initial NRM. A pilot study comparing alternating field (AF) with thermal cleaning on sister samples from the same core showed a very low coercivity, low unblocking temperature (under 250°C) component with steep inclinations before bedding correction (Figure 7). This component with very low unblocking temperatures (150°C to 250°C) may be viscous (carrying a PDF or laboratory direction), and its mineral phase is uncertain. It may be due to superparamagnetic, very fine-grained ("pigment") hematite. Further thermal handling was needed to define the characteristic remanent magnetization (ChRM). After removal of the goethite and viscous components, intensities in the range 10^{-5} to 10^{-6} A/m remained. With continued stepwise heating, the samples often reached the noise level of the cryogenic magnetometer by 400°C. Therefore progressive 25°C steps between 250°C and 400°C were selected for routine demagnetization.

The magnetization was, in general, very weak. Therefore extra care was taken with the individual measurements. Each sample measurement consisted of replicate observations in six positions with a three-axis, computer-assisted ScT cryogenic magnetometer. Each individual observation measurement averaged 20 samplings of each of the three axial components. A statistical parameter allowed immediate rejection of poorly defined vectors on the basis of large values for the standard deviation of angular or intensity deviations [Lowrie et al., 1980].

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TABLE 1. Paleomagnetic Site Statistics

Site	Age	Azimuth/Dip	M/U/R	In Situ Declination	In Situ Inclination	In Situ k_1	In Situ α_{95}	Corrected Declination	Corrected Inclination	Corrected k_2	Corrected α_{95}
					Serres de Llevant						
MJ01	Bathonian	230/60	24/22/0	57.9	2.1	28.3	5.9	61.7	53.6	36.5	5.2
MJ02	Callovian	225/60	23/18/2	31.7	0.3	21.8	7.6	27.4	51.5	33.3	6.1
MJ03	U.Tithon.-M.Val.	220/25	19/17/4	44.6	25.6	14.8	9.6	46.2	45.7	15.7	9.3
MJ04	L. Berriasian	235/40	12/0/0	too weak to measure							
MJ05	U.Tithon.-M.Berr.	210/60	11/0/0	too weak to measure							
MJ13	U. Tithonian	255/50	20/19/2	46.5	3.5	8.8	12.0	43.3	43.3	7.2	13.4
MJ14	L. Bajocian	205/55	18/13/3	51.4	0.7	11.1	13.0	56.2	40.4	10.1	13.7
Mean A (five sites)				46.5	6.4	31.5	13.8	47.1	47.5	62.4	9.8
					Serra de Tramuntana						
MJ06	M.-U. Jurassic	165/20	19/18/0	14.9	44.1	12.8	10.0	32.9	48.1	24.6	7.1
MJ07	Bajocian	127/05	19/14/0	11.9	49.7	30.6	7.3	14.2	42.7	41.1	6.3
MJ08	Bajocian	120/15	21/12/1	27.5	50.6	13.1	12.4	44.3	47.0	15.8	11.3
MJ09	Bajocian	120/-85	12/12/0	80.8	73.2	7.8	16.6	183.3	-3.4	8.0	16.4
MJ10	U. Jurassic	135/20	21/16/8	23.7	61.0	11.8	11.2	48.8	50.2	26.6	7.3
MJ11	Bajocian	150/40	18/13/0	16.2	26.6	11.3	12.9	43.0	50.0	12.6	12.1
Mean (six sites, excluding MJ09)				18.3	46.6	37.1	12.7	36.2	48.3	66.5	9.5
Mean B (five sites, excluding MJ08 and MJ09)				16.2	45.4	30.7	16.8	34.1	48.5	55.4	12.4
					Serres de Llevant and Serra de Tramuntana combined						
Mean of mean A and mean B				34.0	26.7	6.0	44.0	40.7	48.2	171.8	19.2
Mean of ten sites (excluding MJ09)				35.0	27.2	8.7	17.4	41.7	48.0	62.2	6.2
Final mean (nine sites, excluding MJ08 and MJ09)				35.6	24.4	8.5	18.8	41.4	48.1	55.5	7.0

Key to abbreviations: M is number of samples measured, U is number of samples used in statistics, and R is number of reversed samples.

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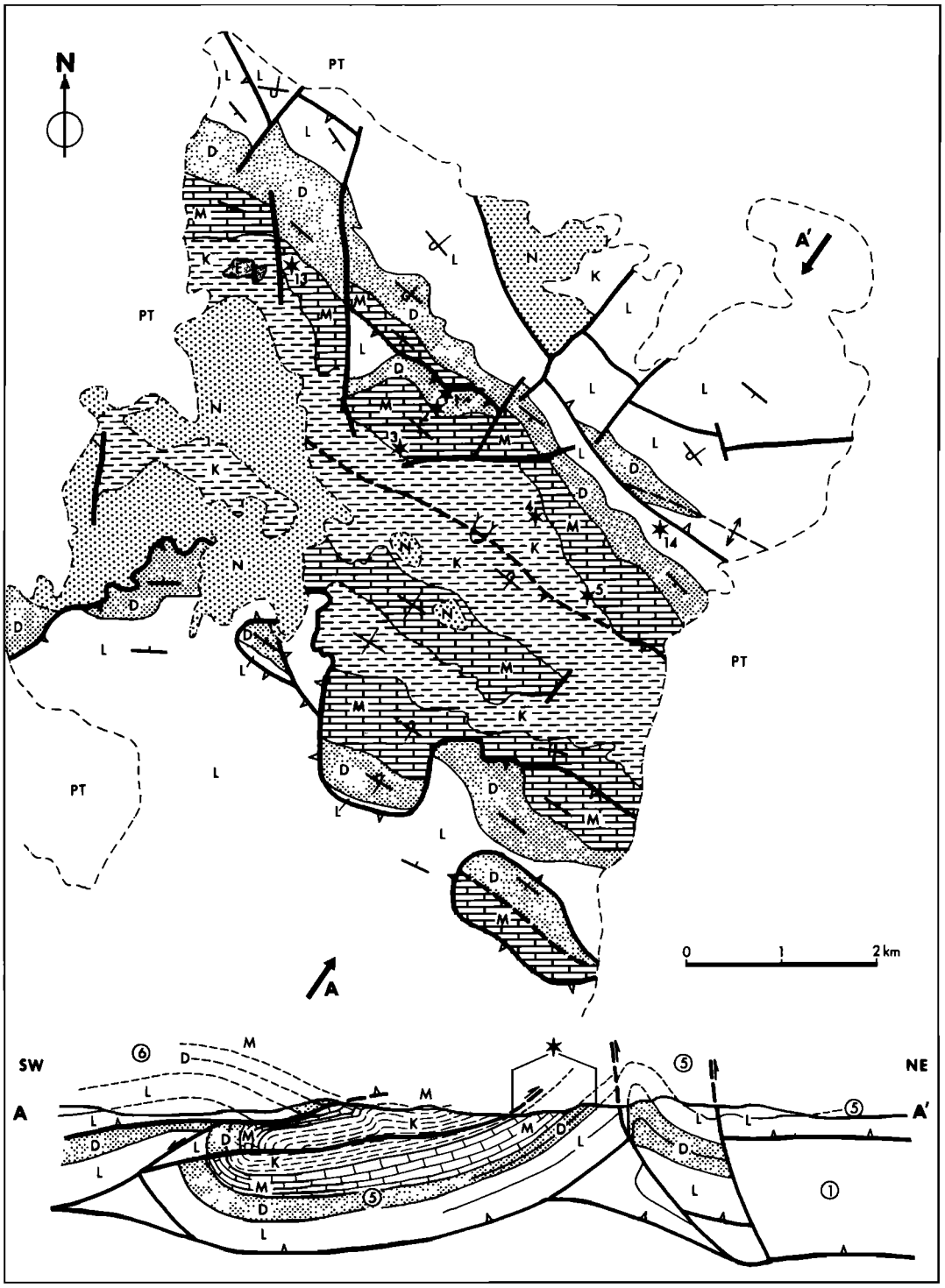


Fig. 4. Paleomagnetic site locations in the Serres de Llevant. Geological map of the part outlined in Figure 2 and cross section (A-A'). Map is simplified from Sàbat [1986]. L, Lower Jurassic; D, Middle Jurassic; M, Upper Jurassic; E, Eocene; N, lower Miocene; PT, upper Miocene. Paleomagnetic sites are located along the normal limb of a recumbant syncline 5 km long and trending NW-SE. Sampled sites are marked by stars (1 is MB01, 2 is MB02, 3 is MJ03, 4 is MJ04, 5 is MJ05, 13 is MJ13, 14 is MJ14. In the cross sections, 1, 5, and 6 refer to thrust sheets defined by Sàbat [1986].

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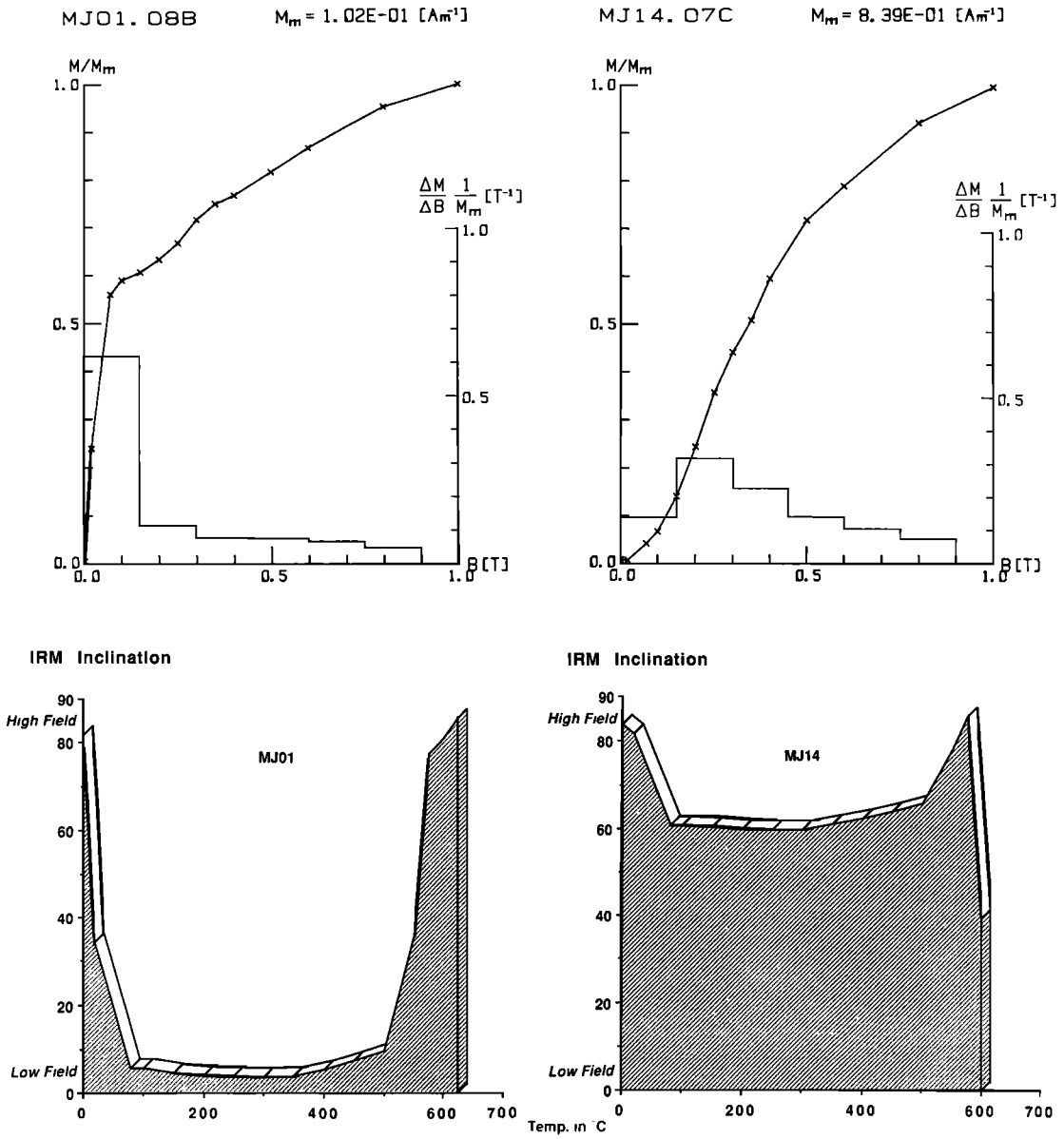


Fig. 5. Examples of IRM behavior. Top: curves of isothermal remanent magnetization (IRM) acquisition. Bottom: the respective plots of the angle ("IRM inclination") between the magnetization and the axial sample direction during stepwise thermal demagnetization after application of a second, perpendicular (axial), low-field IRM component.

Analysis of the NRM demagnetization

Straight line segments on horizontal and vertical projections of a Zijderveld plot of the demagnetization of a typical sample (Figure 7) show that nonoverlapping unblocking temperature spectra characterize the remanent component being removed during demagnetization above 250°C. The vectors associated with the temperature steps that represent the best straight line segments were analyzed by linear regression or principle component (PCA) methods [Kirschvink, 1980]. In cases where no significant directional change occurred before the sample reached the resolution limit we computed the mean direction of the best continuous set of demagnetization steps. ChRM components determined by the PCA method were similar to those determined by calculating the mean direction on samples which exhibited directional grouping but poorly defined straight line segments. Although the two-component IRM demagnetization shows the presence of both magnetite and microcrystalline hematite, the NRM results show no

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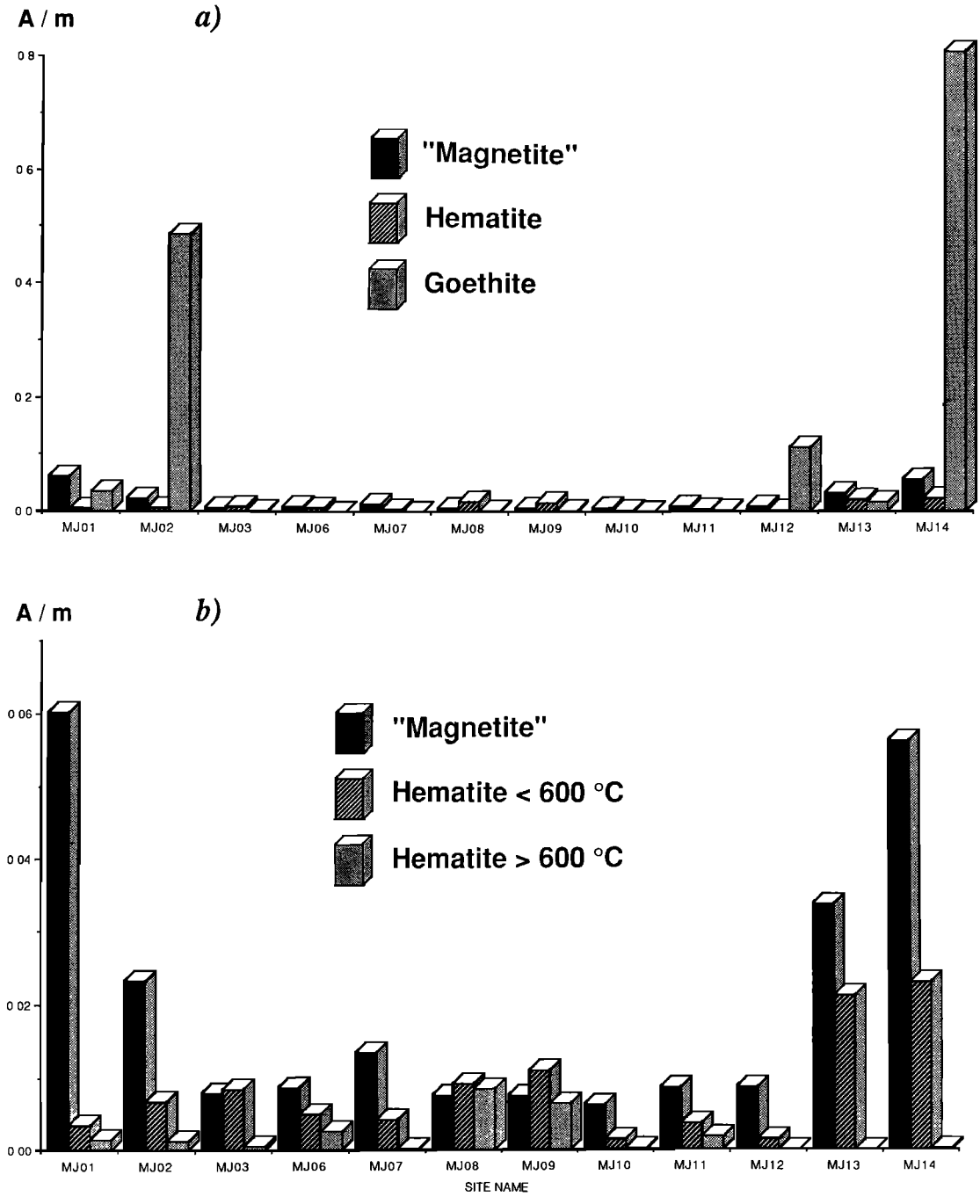


Fig. 6. A semiquantitative comparison of relative amounts of ferromagnetic phases on the basis of the two-component IRM experiment. (a) Including the goethite contribution. Note the dominance of goethite in two of the samples. (b) After removal of the goethite contribution (accomplished by heating above 100°C).

significant directional changes in the range 250°C to 400°C. Since the spontaneous magnetization of magnetite (4.8×10^5 A/m) greatly exceeds that of hematite ($2\text{-}3 \times 10^3$ A/m), it probably dominates much of the ChRM component below 500°C. In a few samples the demagnetization trend appears to overshoot the origin. This may be due to large scatter at very low intensities or to the presence of a higher unblocking temperature component (i.e. hematite). As can be seen from

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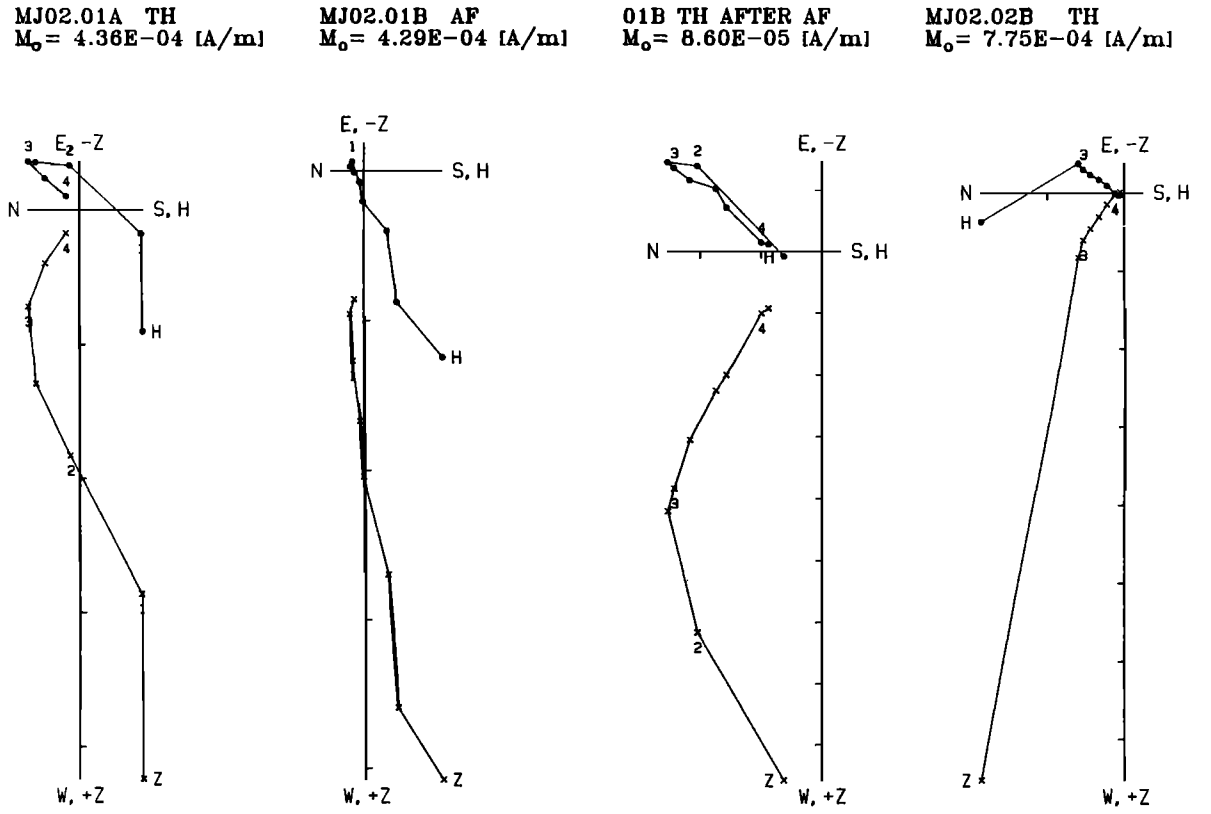


Fig. 7. NRM behavior: Zijderveld plots of the horizontal (H) and vertical (Z) components of alternating field (AF) and thermal (TH) demagnetization. (a) Initial thermal cleaning of a typical specimen. Numbers indicate thermal steps in 100°C. b) AF cleaning of a specimen from the same core as Figure 7a; number 1 indicates AF field of 10 mT. (c) Continued thermal demagnetization of same sample as in Figure 7b. This plot shows that thermal cleaning is necessary to remove the initial overprinting component. The remaining component is the same as in Figure 7a and is identified as the ChRM component. (d) Example of results of the routine thermal demagnetization procedure of a relatively strong sample from the same site as Figures 7a and 7b. The same ChRM component is removed between temperatures of 250°C and 450°C.

Figures 7c and 7d, this component has a slightly shallower and more northerly direction and may be a (more) primary component. However, this component had, except in a few cases, too low an intensity to isolate, and it was impossible to define a consistent direction. We therefore consider the component isolated from 250°C to 450°C to represent the ChRM for each sample, without assuming it was the original (Jurassic) magnetization. The discussion of the results below will show that the question of whether the magnetization is actually primary (i.e., Jurassic) or not has no influence on our conclusions.

Structural corrections to the ChRM data

To eliminate any possible effect due to "apparent rotations" [Macdonald, 1980] the strike and plunge of fold axes were calculated from the measured bedding or map data at each site. These structural data then allowed us to correct the ChRM directions for folding and subhorizontal plunge of the fold axes where this may have had an influence on the structural corrections to the NRM. The ChRM directions from each sample were first rotated to place the bedding tilt onto the great circle which represented the mean dip direction. These were then unfolded using the bedding tilt and then rotated to horizontal according to the plunge of the fold axis. This unfolding order is non-commutative, we assumed that the beds were folded before they were tilted. We do not know that this was actually the case. However, in test cases with the reversed order (first tilt, then fold), the results were very similar to the fold-tilt order. This may be due to the particular orientation of the magnetization and structures on Mallorca; other locations may show significant differences. Since the fold axes plunged, in general, only slightly, the ChRM directions calculated by this correction procedure did not differ appreciably from those corrected by the usual methods.

Paleomagnetic directions

General characteristics. In most cases each core yielded two samples. Because of the generally weak magnetization, both

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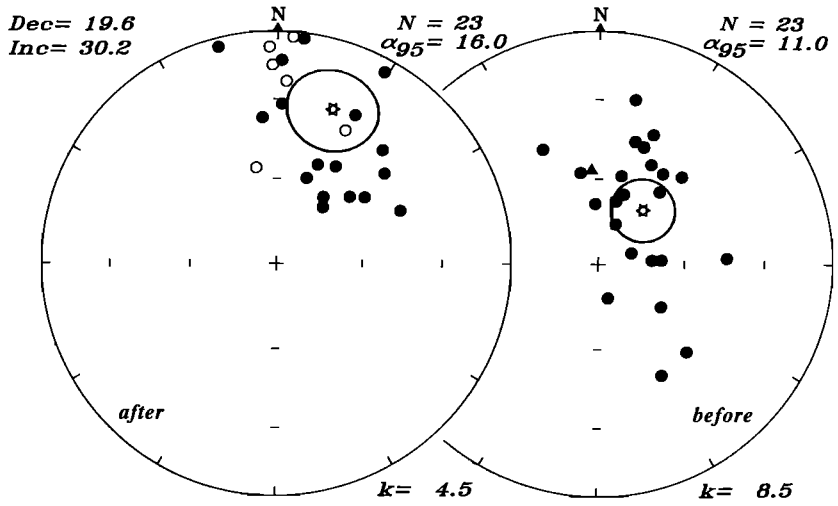


Fig. 8. Stereographic, lower-hemisphere, equal-area projection plots of ChRM vectors from two sites in the Serra de Tramuntana (MJ08 and MJ09) which were used in a fold test. Each point represents the ChRM direction of one specimen as defined by averaging closely grouped thermal demagnetization vectors, principle component analysis, or linear regression of straight line segments of thermal demagnetization. Open circles are reversed, solid circles are normal directions. After tectonic (tilt and fold axis) correction, is "after" and "before" is without tectonic correction. In the upper right-hand corner of each plot are the mean declinations and inclinations (in this plot for the two sites combined) along with the radius of the 95% circle of confidence and the value of the precision parameter "k". Confidence ellipses for the individual sites are shown in Figure10. A triangle marks the present day field direction.

samples from each core were measured and included in the results (see Table 1). Two sites were unmeasurable because of very low intensity of magnetization (MJ04 and MJ05). In the other sites, both normal and reversed polarities were found, but normal polarities dominated. Five sites exhibited no reversed directions. Four of these could be confined to the Bathonian or Bajocian, times of rapid polarity reversals [Steiner, 1980; Palmer, 1983]. Thus there is some doubt as to whether these directions were acquired in the Middle Jurassic, and this will be discussed in more detail later. However, mean directions before bedding correction had shallow inclinations and differed significantly from the PDF direction (see Table 1). Cylindricity of the folds near our sites should correspond to minor declination differences between sites from the same area. This is not always the case. Declination differences within the same area were recorded (see Table 1). Some of these differences might be accounted for by structural complexity introduced by local faults or minor structures. Through detailed geologic mapping we tried to avoid influences of these local structures. However, where detailed small-scale observations were possible, the kinematic indicators were ambiguous, and it was not possible to utilize them for further corrections.

Serra de Tramuntana, fold test. Two sites in the Serra de Tramuntana (MJ08 and MJ09) from the same lithology were located on opposing limbs of a second-order anticline facing NNW, site MJ09 was on the vertical limb very close to the hinge. The mean directions from MJ08 and MJ09 are different from the PDF (Figure 8 and Table 1), but there is a large scatter in the distributions of directions from both sites. To make a fold test, we combined all samples and compared the distribution of the dispersion (or precision) parameter (k) before and after unfolding. The result (k_1/k_2 = 1.89 (note that the k_2/k_1 ratio is inverted to test if the negative result is true); N = 23) fails the fold test at the 90% level but not at the 95% level where it is ambiguous (using table of McElhinney, 1964]). Unfortunately, the hinge of this fold is not exposed, and we cannot ascertain whether it is continuous. The hinge of the adjacent syncline is broken by a small thrust, and thus the negative result of the fold test may be due to the fact that the sampled sites lay on different limbs of a dislocated fold. It is also possible that remagnetization during folding occurred locally near the site at the high-angle limb (MJ09) and not at site MJ08, which was located farther from the fold hinge. These blue-grey and yellowish-beige hemipelagic marly limestones can be distinguished from the other lithologies by their color and also through comparison by ferromagnetic mineralogy on the basis of IRM acquisition curves and two-component IRM demagnetization (see Figure 6). The site mean direction from MJ08 and MJ09 were excluded from further consideration.

Serra de Tramuntana, ChRM results. The remaining four sites (MJ06, MJ07 MJ10, and MJ11) from the Serra de Tramuntana were from pelagic micritic limestones. Their in situ site means, although lying close to the PDF, differed significantly from it (Table 1 and Figure 9). For two individual sites (MJ06 and MJ10) the improvement at the site level of the distribution after tectonic correction is significant at the 95% confidence level (using table of McElhinney, [1964]) (see Table 1).

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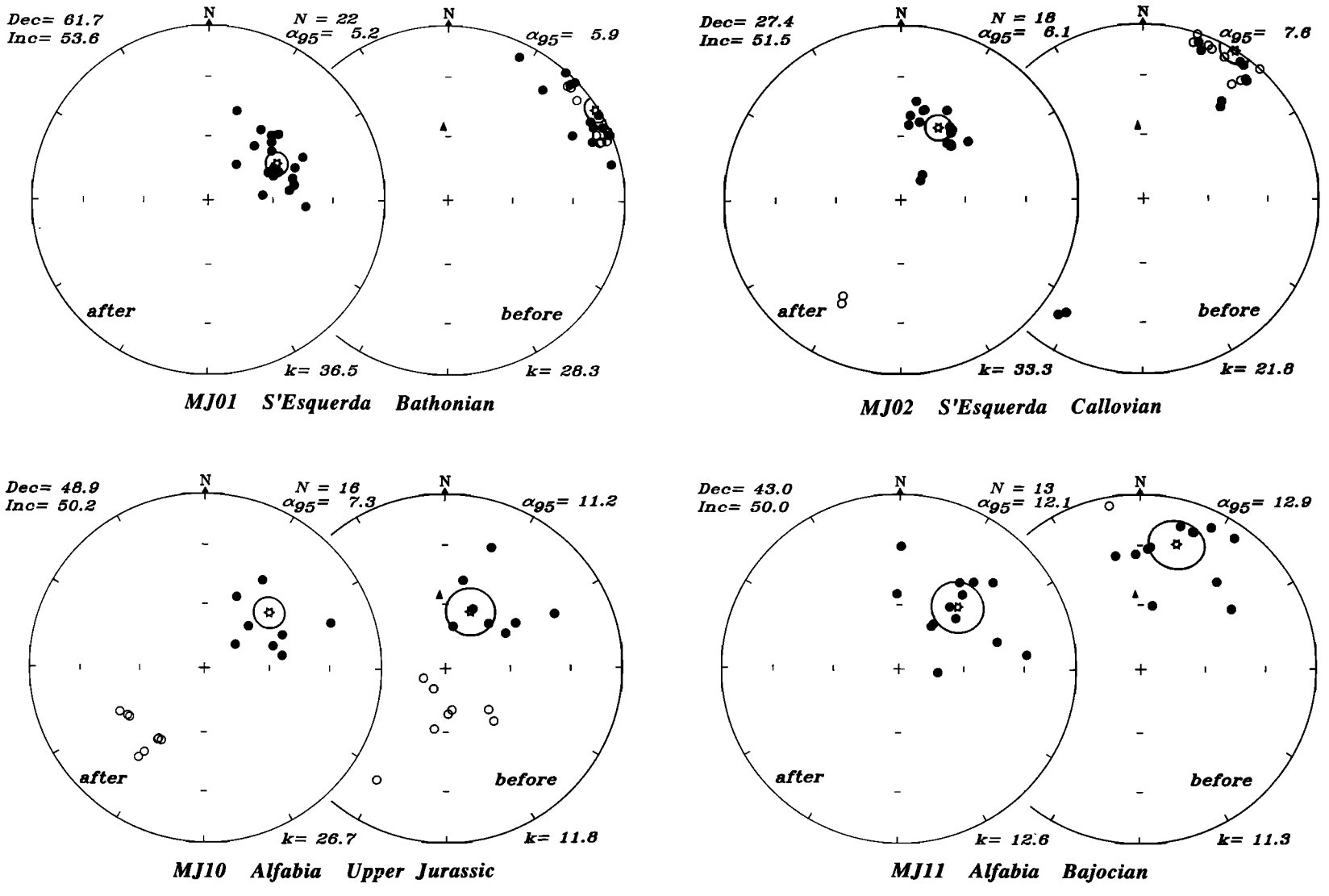
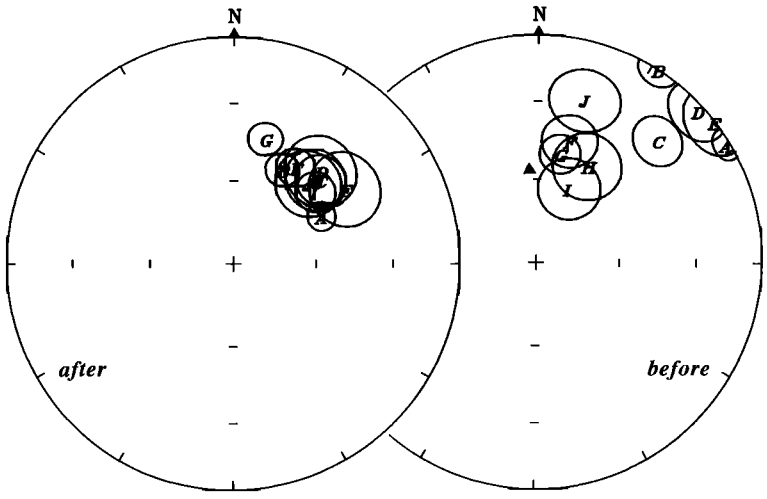


Fig. 9. Stereographic, lower-hemisphere, equal-area projection plots of ChRM vectors from two sites in the Serres de Llevant (MJ01 and MJ02) and two sites from the Serra de Tramuntana (MJ10 and MJ11). Details as for Figure 8.

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Serres de Llevant + Serra de Tramuntana site means

Fig. 10. Stereographic, lower-hemisphere, equal-area projection plots of the 95% circles of confidence of site means from the Serres de Llevant (A is MJ01, B is MJ02, C is MJ03, D is MJ13, E is MJ14,) and Serra de Tramuntana (F is MJ06, G is MJ07, H is MJ08, I is MJ10, J is MJ11). Note that site MJ08 (H) is included in this plot for comparison but not in the final statistical results nor in Figure 11. The overturned site MJ09 is not shown here. The improvement of the distribution (k increase from 8.5 to 55.5, see Figure 11 and Table 1) is the main structural result of this paper: the paleomagnetic data show no evidence for differential rotation between the Serres de Llevant and the Serra de Tramuntana.

Taken together to define a "thrust mean", these four sites yield, after structural corrections, a declination of 34° and inclination of 48°. The radius of the circle of 95% confidence is reduced after tectonic corrections from 17° to 12°, and the value of the dispersion parameter k is increased from 31 to 55 (see Table 1 and Figure 10). Although an improvement, this increase is not statistically significant at the 95% level. Weak magnetization of individual samples and presence of local structural distortions result in large k values and lessen the usefulness of this statistical test. (As an example, if we exclude site MJ07, this same test yields an improvement which is significant at the 95% level.) However, where they are significant, the statistical results support a prefolding origin for the magnetization. A comparison of site means taken collectively for both thrust sheets together and the means calculated for both thrust sheets before and after tilt correction yield additional levels of statistical tests with significant results, and these will be discussed below.

Serres de Llevant. ChRM directions from the Serres de Llevant are statistically far from the PDF before tectonic correction (Table 1 and Figure 9). With tectonic correction, site means from the Serres de Llevant group well (k = 62) and the α_{95} circles of confidence overlap. However, although improved by a factor of nearly 2, the change in the dispersion parameter is not significant at the 95% level. As for the Serra de Tramuntana, this test does not support either prefolding or postfolding magnetization. The basic problem is that we have sampled from a limited range of structural positions. Other causes are weak magnetization and local structural distortions. If these distortions account for the scatter of paleomagnetic directions, then the similar k values before and after structural correction merely say that we have not adequately corrected for the local distortions. However, field study by Casas and Sàbat [1987] has shown that it is problematic to correct for the scatter introduced by these local, small-scale structural distortions. The scatter may indeed indicate that the directions, being affected by the tectonization, are of pretectonic origin. In spite of the fact that the assemblage cannot be used to unambiguously test the age of magnetization, we can still take these five site means to create a mean direction representative of the Serres de Llevant thrust sheet. This gives a declination of 47° and an inclination of 48° (Table 1). Similar to the results from the Serra de Tramuntana, magnetization in the PDF can be discounted, but a test of the improvement of the dispersion parameter relative to folding or tilting is statistically insignificant at the 95% level.

Conclusions, combined data sets. In contrast to the calculation of the mean for each thrust sheet, when all site means are considered together (illustrated in Figures 10 and 11) there is a significant improvement of the distribution even at the 99% confidence level (k increases from 9 to 56; N = 9, see Figure 11 and Table 1). Another test on a higher statistical level is to first calculate the mean for each thrust sheet and compare these two. This result is shown in Figure 12, and the improvement in the dispersion parameter (k increases from 6 to 172; N = 2) is significant at the 95% level. These results imply that the sites from both thrust sheets were magnetized prior to folding.

Figure 10 also illustrates another important result: the circles of 95% confidence are relatively scattered before correction and except for site "G" (MJ07) they overlap after correction. In

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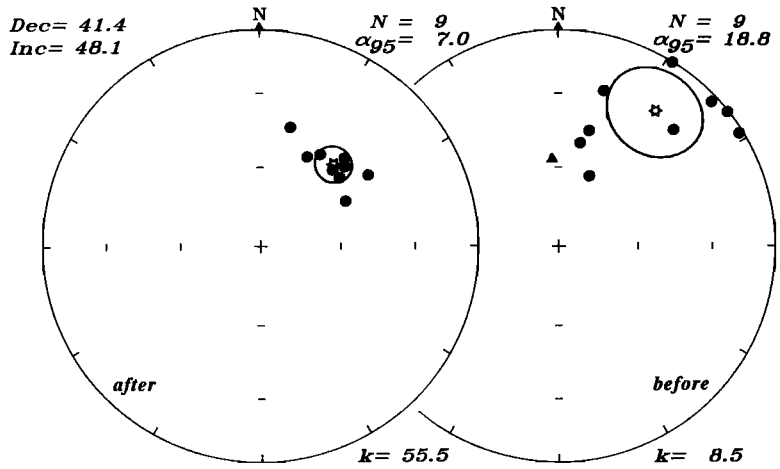


Fig. 11. Stereographic lower-hemisphere, equal-area projection plots of site means from the Serres de Llevant and Serra de Tramuntana combined. Details as for Figure 8. Sites MJ08 and MJ09 are not included in this plot. Note the improvement in the "k" and "α95" parameters after bedding correction. The declination of 41.4° identifies the significant tectonic result of this paper: the paleomagnetic data imply considerable clockwise rotation since the time of acquisition of magnetization.

addition, as shown in Figure 13, the confidence circles for the "thrust sheet means" also overlap after tilt correction. Thus the mean directions are statistically different before correction but statistically the same after structural correction. The simplest interpretation is that the thrust sheets were magnetized before folding in the same orientation and throughout folding were not subjected to differential rotation. Other interpretations would require improbably complex and unlikely combinations of pre-folding magnetization positions and postfolding thrust-sheet contortions.

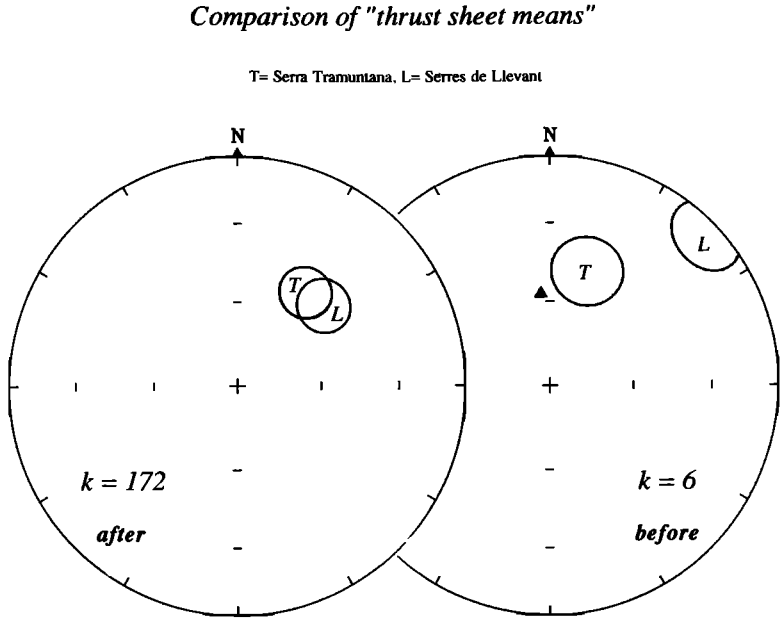


Fig. 12. Stereographic, lower-hemisphere, equal-area projection plots of "thrust sheet means" from the Serres de Llevant and Serra de Tramuntana before and after tilt correction. Note the improvement in the dispersion parameter "k". Black triangle marks the PDF location.

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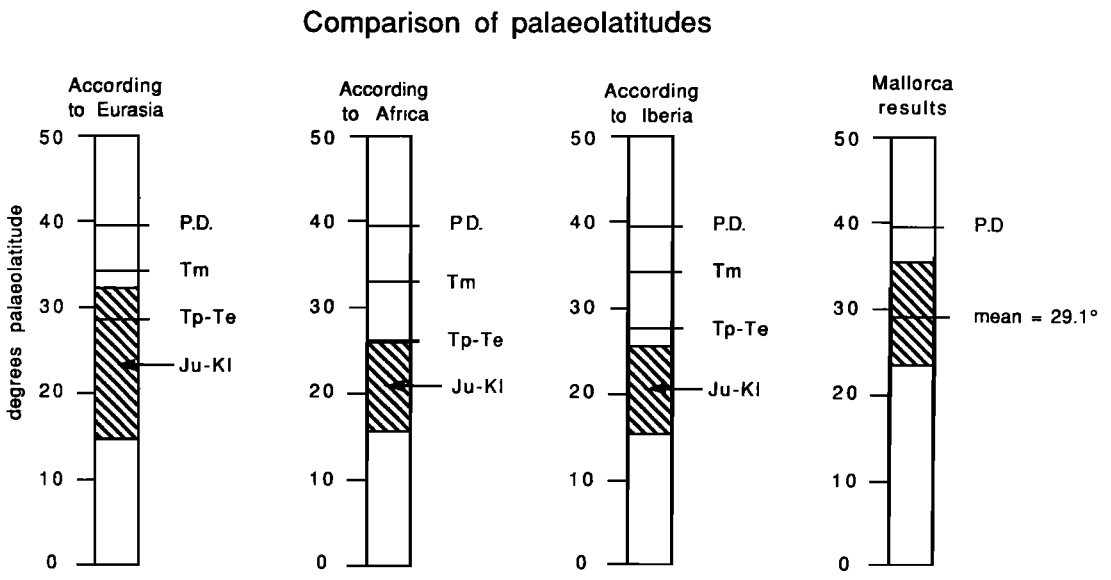


Fig. 13. Comparison of paleolatitudes for a site on Mallorca calculated from apparent pole wander curves for Eurasia, Africa, and Iberia (from kinematics), given in Westphal et al. [1986]. Declinations and inclinations are listed in Table 2. Hatched areas indicate the range of paleolatitudes using the respective values of α_{95} calculated from the source data. The age best corresponding to the paleolatitude of the measured Mallorcan data is Paleocene-Eocene. However, the errors are too great to place much weight on this correlation.

Kinematic significance of the paleomagnetic results

The kinematic consequences of these paleomagnetic results become clear once the structural situation is reviewed. The structural data show that both thrust sheets have been transported with the same transport direction but that the Serres de Llevant thrust sheet has structures genetically related to oblique-lateral ramps, where field study has deduced large amounts (about 40 km) of displacement [Sàbat, 1986]. The Serra de Tramuntana thrust sheet, in contrast, has structures genetically associated with a frontal ramp. During displacement some relative rotation between these two thrust sheets could be expected after considering the large difference between their boundary structures. Rotational components could have arisen from friction along oblique ramps due to the large magnitude of displacement. However, the paleomagnetic data imply that both thrust sheets were transported by pure translation with no differential rotation (or at least with the same amount of rotation). Our results suggest that thrust sheets can be moved along oblique-lateral ramps and shortened in a direction nearly perpendicular to transport direction without producing significant rotations.

Timing of acquisition of the ChRM

The statistically significant paleomagnetic results imply that the directions predate Oligo-Miocene folding and thrusting (with the exception of site MJ09 which may have a synfolding origin). However there is a conspicuous absence of reversals in four out of six Bathonian-Bajocian sites (three out of four excluding MJ08 and MJ09), which cast doubts on whether these ChRM directions are of Jurassic age. A meaningful comparison of the inclination results with other data is hampered by the large variation (from 30° to 45°) in the published Iberian paleomagnetic data for the Jurassic [Vandenberg 1980; Schott et al., 1981; Ogg et al., 1984; Steiner et al., 1986]. The mean inclination from the combined set of nine Mallorca Jurassic sites is 48°, slightly steeper than the published range. However, it is not significantly different considering the 7° radius of the α_{95}. This inclination value implies a paleolatitude of 26°-32°, which is also slightly higher than the 20°-23° for the Balearics shown in the palinspastic geological maps for the Middle and Upper Jurassic [e.g., Dercourt et al., 1986].

Another comparison is between paleolatitudes calculated using the Eurasian, African, or Iberian apparent polar wander paths [Westphal et al., 1986], which is shown in Figure 13. Comparison of the paleolatitudes calculated from our data for Mallorca with those for the Middle Jurassic to Upper Jurassic-Lower Cretaceous (hatched areas in Figure 13) and those for the Paleocene-Eocene clearly show that, for all apparent polar wander paths, the most likely dating would be early Cenozoic. Paleomagnetic sampling in NE Spain [Parés, 1988, Parés et al., 1989] has evidenced widespread remagnetization in most of the Mesozoic limestone lithologies. Thus we suggest that remagnetization is likely.

One possible process for this suspected remagnetization is secondary magnetization associated with weathering. Between the Upper Cretaceous and lower Eocene there was a period of uplift and emersion with weathering and partial erosion of the Jurassic limestones, during which time they may have acquired their ChRM. In general, on Mallorca a gap exists in the stratigraphic record between lower Cretaceous pelagic marls and middle or upper Eocene detritic limestones. In most cases the

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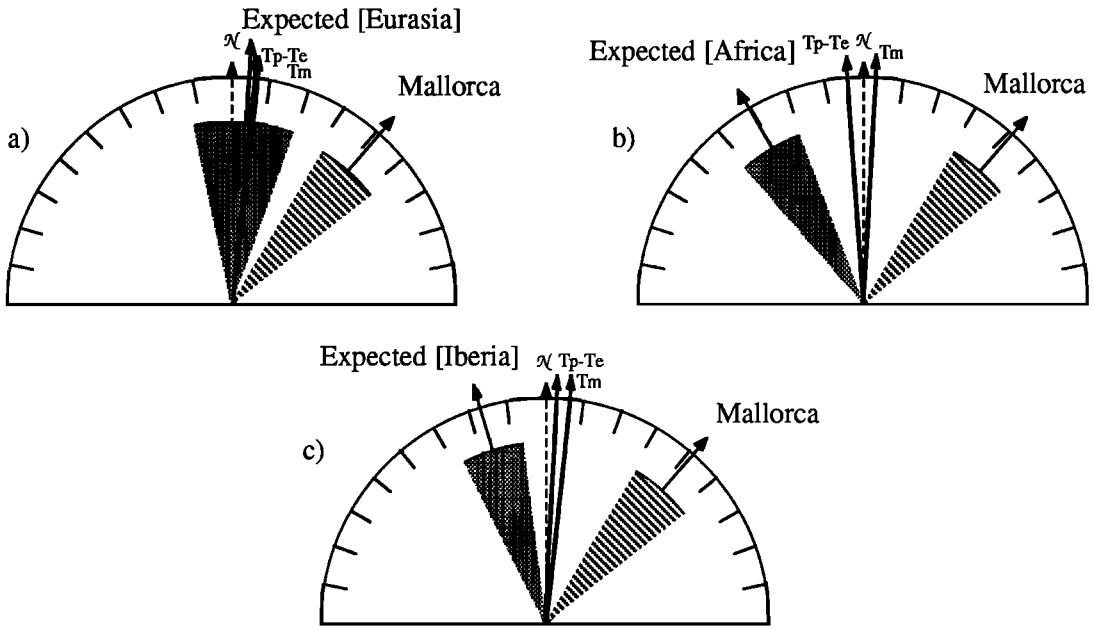


Fig. 14. Comparison of the expected declinations calculated for a site located on Mallorca using the data for Eurasia, Africa, and Iberia of Westphal et al. [1986]. Shaded pie-shaped regions indicate the uncertainty in declination (ΔD) as listed in Table 2. In all cases the measured data show large clockwise deviations. Do not see text for explanations because we do not have any yet; regional extension in the Neogene most likely played a significant but not unique role.

contact is paraconformable. Upper Cretaceous pelagic limestones exist in a few localities which escaped erosion. Paleocene emersion then eroded the Upper Cretaceous part of the Mesozoic cover. Further erosion is evidenced by Eocene marine conglomerates which contain pebbles of Lower, Middle, and Upper Jurassic limestones [Hollister, 1934; Bourrouilh, 1973; Stoeckinger, 1976a,b]. The Jurassic rocks may have acquired their ChRM during this emersion from Paleocene to lower Eocene. At present we have neither enough data nor an unambiguous fold test to determine if the ChRM is Jurassic and carried by magnetite or remagnetized and carried by fine-grained, secondary hematite with an early Tertiary remagnetization age. Secondary magnetization in an (unknown) uplifted and slightly tilted position could also explain the scatter which does not significantly improve with tectonic correction. For the conclusions presented in this paper it is not critical if magnetization occurred in the Jurassic or Early Tertiary. The available geologic and paleomagnetic data indicate that if remagnetization occurred, then it took place prior to Oligocene-Miocene folding and thrusting and most probably in the Paleocene.

THE REGIONAL TECTONIC PROBLEM

Declinations from all site means show clockwise rotation after tectonic correction (Figure 11). Figure 14 compares the mean declination of nine sites on Mallorca with expected Jurassic and Tertiary declinations recalculated for the present location of Mallorca using Eurasian, African, or Iberian apparent polar wander paths (Table 2) listed by Westphal et al. [1986]. Assuming Paleocene-Eocene remagnetization with Mallorca attached to Eurasia (expected declination for a site on Mallorca = 6°), there has been 35° ± 10° clockwise rotation of both thrust sheets. All other reconstructions yield larger net clockwise rotations.

Recent results [Freeman et al., 1988] from autochthonous, post-orogenic carbonates from Mallorca and Menorca (the northernmost Balearic island) indicate that 25° ± 15° clockwise rotation has occurred since the upper Miocene. These results imply basement involvement rather than just movement of the allochthonous cover. Although we do not know enough to determine the mechanism for this rotation, there is ample evidence for regional Neogene extension that could have produced clockwise rotations. The Valencia Trough was formed by Neogene extension [Biju-Duval et al., 1978; Vegas et al., 1980]. However, the opening of the Valencia Trough cannot account for more than 10° of clockwise rotation. It is conceivable that processes associated with the opening of the Provençal Basin (Gulf of Lions) and the (counterclockwise) rotation of Corsica and Sardinia in Oligo-Miocene times may have produced southeastward movement of Mallorca, with a clockwise rotational component.

On the other hand, there are local features which indicate strike-slip activity. Younger faults on Mallorca cut Oligocene to Langhian (middle Miocene) folds and thrusts. These Neogene faults control the basin shape and depocenters of upper Miocene, Pliocene, and Quaternary sediments [Del Olmo and Alvaro, 1984]. Some of these faults reveal major strike-slip movement shown by horizontal striations [Pomar et al., 1983].

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TABLE 2. Expected directions for Mallorca

Age	Eurasia		Africa		Iberia	
	Declination	Inclination	Declination	Inclination	Declination	Inclination
Miocene	6	54	3	52	6	54
Paleocene-Eocene	6	48	356	44	3	47
J_U-K_L	1	37	328	35	339	36
J_U	7	35	330	33	345	34
J_M	11	41	329	42	347	47
Mean J_M-K_L	4	40 (ΔD=16°)	329	37 (ΔD=10°)	343	37 (ΔD=10°)

Expected and observed declinations calculated for a site located on Mallorca (longitude 3.0°E and latitude 39.6°N) using selected Eurasian and African pole locations and a polar wander curve for Iberia reconstructed according to kinematics (data from Westphal et al. [1986]). K_L, Lower Cretaceous; J_U, Upper Jurassic; J_M, Middle Jurassic; ΔD, uncertainty in declination [ΔD=arcsin (sin α_{95}/cos Inclination)]. Westphal et al. [1986] warn that the Upper Jurassic (J_U) pole directions are especially poorly defined for Eurasia and Africa.

These faults also have significant vertical throws, which give Mallorca a block-faulted structure and contribute to determining the polygonal contours of the island (Figure 2). They are associated with regional transtensive conditions which have prevailed since the middle Miocene [Tapponnier, 1977; Vegas et al., 1980].

One, or a combination, of these mechanisms may have produced clockwise rotation since the upper Miocene. The remaining approximately 20° of clockwise rotation would then have been acquired between the timing of magnetization and the upper Miocene. This rotation could also have been produced by earlier phases of the previously mentioned mechanisms, or it could be related to Oligo-Miocene compression that produced the thrusting on Mallorca.

Unfortunately, we do not have enough data or knowledge at present to substantiate which of these mechanisms operated or to produce more definite models. To obtain more paleomagnetic data from the western Mediterranean, we have undertaken a comprehensive sampling of a wide range of lithologies and ages from the Balearic Islands, the Catalan Coastal Range [Parés, 1988; Parés et al., 1988, 1989], and the Subbetics [Osete et al., 1988]. These results and their tectonic implications are now being evaluated.

SUMMARY

These are the first paleomagnetic data with tectonic implications from a paleomagnetically unknown area of the western Mediterranean. Although the exact timing of the acquisition of magnetization is uncertain, the paleomagnetic data indicate that it was earlier than thrusting and folding during the Oligocene-Miocene orogenic event. Applied to the problem of differential rotation of thrust sheets on Mallorca, the similar ChRM directions from the Serra de Tramuntana and the Serres de Llevant indicate no differential rotation. This implies that thrust sheets can be moved along oblique-lateral ramps and shortened in a direction nearly perpendicular to transport direction without producing significant rotations. In the regional tectonic situation of the western Mediterranean the paleomagnetic declinations indicate 35° ± 10° clockwise rotation since the timing of magnetization, which must have been prior to the Oligo-Miocene. This rotation should be taken into account when structural data from pre-Miocene rocks are used for regional tectonic interpretation.

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J.-M. Fontboté and F. Sàbat, Departament de Geologia Dinamica, Geofisica i Paleontologia, Facultat de Geologia, Zona Universitaria de Pedralbes, 08028 Barcelona, Spain.

R. Freeman and W. Lowrie, Institut für Geophysik, ETH-Hönggerberg, 8093 Zürich, Switzerland.

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